

Which Short Run? Whose Money? Closure Choice in Input–Output Models of the Socio-Ecological Transformation*

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Abstract

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Input-output models are a widely used tool for estimating the economic effects associated with societies moving towards carbon-neutrality or, more fundamentally, aiming to achieve a socio-ecological transformation. While this practice documents that such models are perceived as conceptually useful and empirically relevant, such approaches often take input-output models as a supposedly neutral analytical device. Against this background, this paper revisits how paradigmatic perspectives matter for input-output modeling. It shows that such perspectives and the associated analytical differences have a substantial impact on estimated effects at various levels. This observation indicates how modern applications of established models are actually haunted by ghosts from the past that reflect long-standing debates, and this has impact on the interpretation of a great variety of published results. Using the case of an archetypal green transformation policy, the decarbonization of the German housing sector, we show how different axiomatic setups imprint on estimates of the economic effects of ecologically motivated transformation efforts. By doing so, we hope to provide applied researchers with better guidance for reflecting on how the diversity of available interpretations of input-output models impacts

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on their problem at hand.

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JEL-Codes: C67, C68, D57, E24

1 Introduction

Input-output models are an established tool for estimating the economic effects of structural policy choices related to public investment, industrial policy or labor market trends. Against this background it does not come as a surprise that input-output models are also widely used to estimate the economic effects of additional investment undertaken in the course of a socio-ecological transformation aiming to render social provisioning processes carbon-neutral (SOURCES). However, it is well known that the results obtained from such models depend crucially on the underlying axiomatic setup. Variations in outcomes emerging from variations of the chosen setup are seldom reported in the literature on transformation efforts that resorts to input-output modeling. Moreover, and more importantly, little attention is paid to the fact that the axiomatic setup of input-output models reflects economically important contextual arguments. In this paper we recapitulate that substantial variation in outcomes is due to a specific part of the axiomatic setup, namely the different closure choices. For research on socio-ecological transformation efforts it is important to recognize that such closure choices are not mere technical decisions, but carry substantial economic content; they capture – or presuppose – certain contextual arguments, assumptions about the current state of the economy, the treatment of the future and the exact form in which some policy proposal is executed in practice.

Closure choices carry substantial economic content as they make (implicit) assumptions about the context and conditions under which the suggested policy is implemented. This observation resonates with arguments on the importance of socio-historical and institutional context in economics (SOURCES) as neatly summarized by Keynes's quip that "economics is a science of thinking in terms of models joined to the art of choosing models which are relevant to the contemporary world" (SOURCE). In this spirit our argument is also grounded in the observation that closure choices often reflect traditional divides in economic theory and policy so that, in effect, old ghosts haunt modern applications of input-output models, which have become a main workhorse in modeling socio-ecological transformation efforts (SOURCE??).

It is well known that such paradigmatic cleavages matter greatly for input-output analysis: while the question whether or to what relative extent output or prices react to a change in final demand (scarcity vs. output-adjustment) is most prominent, a series of other conceptual questions come into view that reflect such paradigmatic cleavages. Choices on production technology (Cobb Douglas vs. Leontief), financing (external vs. internal), policy implementation (incentivising private actors

vs. state-led action) or technological change (exogenous vs. endogenous) are all examples of such conceptual questions that have been discussed in the literature on input-output modeling for decades. In this paper we show that such conceptual questions are not merely of academic interest, but have a substantial impact on the estimated effects of socio-ecological transformation efforts. To substantiate and illustrate this claim we use the case of a well-understood green transformation policy, the decarbonization of the German housing sector, and show how different axiomatic setups imprint on final estimates. By doing so, we hope to provide applied researchers with better guidance for reflecting on how the diversity of available interpretations of input-output models impacts on their problem at hand.

Taking our own argument on the importance of paradigmatic cleavages seriously, we aim to summarize some classic questions arising in the history of input-output modeling and indicate how they map onto current efforts in socio-ecological transformation research. We show how the underlying closure choices reflect contextual arguments and can – in terms of input as well as outcomes – be mapped to certain paradigmatic perspectives. By doing so we also show that some combinations of assumptions lead modeling approaches, that are traditionally conceived as competing, to reach similar results. Furthermore, we illuminate which modifications or intermediate approaches might be of interest for researchers trying to credibly model the impact of socio-ecological transformation efforts. In this spirit, we also take some recent developments in mainstream economics into account that move beyond the traditional assumption of aggregate rigidity in terms of fully clearing factor markets, but assume additional rigidities on labor markets.¹ Interestingly, we find that the studied modifications lead to a less optimistic view on the capacities of market adjustments; however, as we will see this leads to an even stronger laissez-faire bias in policy implications as the economic status quo becomes a relatively superior, but still fragile construct, that shall not be disturbed exogenously.²

In what follows we will specifically focus on two types of closures, namely labor market closures and financing closures, which can be mapped on a two-dimensional space. In this space one axis goes from fundamental scarcity of factors (i.e. fully utilized factor markets) to flexible output adjustment (in the spirit of the Keynesian multiplier), which leads to fundamentally different conceptions on the short run. The second axis focuses on financing and goes from the assumption of immediate

¹Related work on rigidities in capital or financial markets also exists, but is beyond the scope of this paper (see, however, (SOME SOURCES)).

²In this case assumptions deviate from the neoclassical baseline, but strengthen the general laissez-faire policy stance associated with its standard closure. A similar, although less pronounced pattern, can be observed for the assumption of a Leontief production function, which similarly decreases flexibility and adaptive capacities of markets.

refinancing (additional expenditures lead to cuts elsewhere) to assuming some form of external financing that relieves the contemporary economy from immediate refinancing needs. These axes are conceptually captured by Table 1 below, which gives a first, simplified representation of the cleavages under study.

[Table 1 about here.]

The table is deliberately kept simple: it collapses the labour-market axis into two poles and is silent on magnitudes. The paper’s central object is the finer grid behind it – a five-by-three closure matrix (five labour-market closures crossed with the three financing regimes above), evaluated on a single, empirically informed green-investment impulse. Running that grid illustrates our main message, namely there is no closure-free multiplier. Every estimate of a transformation programme presupposes a short run and a financing regime, and that choice is not a technicality – it is first-order for the magnitude, and occasionally for the sign, of the result. The matrix then identifies where closure choice is redundant (the rows that coincide exactly mimicking the basic intuition of Table 1) and where it decides the answer.

The remainder of this paper is structured as follows...

2 From transformation research to input–output models

Modest proposals for decarbonization initiatives and large-scale suggestion for fundamental transformational both imply a repurposing of spendnig, either by private agents, the public sector or an interaction of both. Hence, such programmes share a similarity with traditional public spending programs (e.g SOURCE, ZB POLLIN), which also invite to reflect about the economic impacts of such repurposed or additional spending. Classic concepts that come to mind in this context are the Keynesian cross (implicitly building on the idea of external financing), the Haavelmo Theorem (which just states that that state – in contrast to households – spends all his income) or the idea of a “crowding out” of public spending (similar to Haavelmo, but typically with adverse effects due to less investment)

Several contributions in the recent literature (SOURCE; MAZZUCATO AND THE LIKE) exploit this similarity in various ways arguing that public spending is a functional means or necessary requirement for decarbonization and socio-ecological transformation. Aside from such more technocratic takes a series of authors identify poltiical synergies on this basis, where public spending alleviates both, environmetnal degradation as well as social exclusion, and, hence, serves as a foun-

dational keytone for proposals on a just transition. MOreover, analytical approaches to input-output models sometimes emphasize their capability to identify structural dependencies and critical junctures, allowing for assessing the resilience and versatility of economies in times of intense transitions (SOURCE; DOYNE FARMER BOOK).

Finally, as traditonal application of input-output models in the spirit of a simple application of a Leontief inverse assume and output expansion analogous to the Keynesian cross this basic intuition of synergy can also be found on a more analytical level – albeit only when taking, maybe unconsciously, a specific paradigmatic perspective. Given their analytical and conceptual adavantages, their wide applicability and their resonance with contemporary political challenges it does not come as surprise that also many applied studies in transformational research make use of input models to assess the potential impacts of such policies (LONG LIST OF SOURCES).

Our exemplary application goes to the heart of these debates by focusing on the key issue in transformation debates, namely the decarbonization of the residential sector. By taking Germany as an example we focus on a major European economy that is coined by a strong path dependence towards the use of fossil energy sources, especially in the residential sector, but also in energy production, which is key for provoding alternative means of provisioning basic housing services. We start from a concrete estimate for additional investment requirements in Germany to arrive at carbon neutral residential sector till 2050 (Hornykewycz et al., 2025). Such an effort would require to more then double spending on renovaiton and increase annual expenditures on this by about 58 bn on average. By taking this number – and the estimated sectoral distribution of spending needs – as our vantage point, we anchor our illustration in an empirically and politically relevant case that is a valid example for the kind of “repurposing of existing expenditures” that a socio-ecological transformation would entail.

3 Two primitive questions

3.1 A vision of the short run

3.2 A theory of money

4 One kernel, five closures, three financings

4.1 The five labour closures

4.2 The three financing closures

4.3 Why only three economies

4.4 Presentation rule

5 Application: the programme through the matrix

5.1 Aggregate results

5.2 Sectoral results

5.3 The financing wedge

5.4 Robust vs closure-sensitive

6 Beyond demand-only shocks

6.1 The equivalences and their scope

6.2 The sectoral family

6.3 The supply-shock arm

6.4 The no-go corollary and the BF pole

7 Discussion

8 Conclusion

Data availability

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Conflict of interest

The authors report no conflicts of interest associated with the content of this paper.

References

Hornykewycz, A. et al. (2025). “Carbon neutrality in the residential sector: a general toolbox and the case of Germany”. In: *npj Climate Action* 4.1, p. 31.

	Labour-market closure			
	Factor (scarcity)	supply	fixed	Real wage fixed (output ad- justs)
Repurposing consumption (F1)	pure reallocation			the change follows the input– output structure
Refinancing via taxes (F2)	traditional crowding out			balanced-budget (Haavelmo) multiplier, dampened by leakage
External finance (F3)	welfare loss and external debt			endogenous money, external debt and output expansion

Table 1: A first, simplified map of the two closure axes. The full 5×3 matrix follows in Section 5.

A Full proofs

This appendix reproduces, in full, the ex ante equivalence results of the revision (source note: `equivalence.tex`, v4). It states the common kernel, derives the demand-only reduction, proves the exact equivalences of the closure rows and of the sectoral family’s rigid corner, isolates what does differ across the rows, and derives the scope conditions under which the equivalences break.

A.1 Purpose and summary

Reviewer 2’s critique turns on the exact conditions under which the closure rows of the evaluation matrix are or are not equivalent (`docs/DOCS_ASSESSMENT.md`, section 4.2). The executed `matrix_5x3_v10` generation (ADR-0019 together with the ADR-0020 option C endpoint; 33/33 cells, run `ids matrix_5x3-v10-*`) measures two coincidences:

- **ALPHA** $\equiv$ **BETA**. The elastic-supply row ($\eta_s = 0.5$) reports every metric of the full-employment row identically: real GDP, the welfare index, employment, the external account — to all eight printed decimals, with the identity wedge at $\sim 10^{-16}$. A direct comparison of stored solutions measured $\max |\Delta y| = 6.2 \times 10^{-17}$ (session of 2026-09-17). The v6 generation reproduces the v5 ALPHA/BETA cells to 6.7×10^{-16} , so the coincidence survives the endpoint change.
- **GAMMA** $\equiv$ **DELTA**. The Leontief-corner row $((\theta_\delta, \epsilon_\delta, \sigma_\delta) = (10^{-4}, 10^{-4}, 10^{-4}))$ reports every metric of the fixed-wage row at the baseline CES elasticities $(\theta, \epsilon, \sigma) = (0.5, 0.5, 0.9)$ identically, in all three financing columns; the flow-table wedges agree at the residual level ($\sim 10^{-12}$). The analytic `leontief_multiplier` test reproduces the nonlinear DELTA solution with relative error 0.0 (recorded in `docs/DOCS_ASSESSMENT.md` section 4.2).

Sections A.2–A.3 set up the common kernel and derive the demand-only reduction. Section A.4 states and proves the two equivalences and classifies them. Section A.5 records a third exact equivalence, of a different kind: the rigid corner of the sectoral-labour-market family is the $\eta = 0$ endpoint itself, and the family interpolates between the matrix’s two rigidity rows. Section A.6 isolates what *does* differ across the rows (the scalar closure regimes). Section A.7 derives the scope conditions: which model changes break the equivalences — the menu behind the demand-sensitive-price program — and which cannot.

Throughout, “demand-only shock” means the matrix design: a programme bundle $g = G_0 \psi$ ($\sum_i \psi_i = 1$; $G_0 = 40\,300$ EUR m, ADR-0024) entering final demand under one of the three financing closures,

with the supply shock at $A \equiv 1$ (unit productivity).

A.2 The common kernel

Let N be the number of sectors. Prices p_i , wage w , gross output y_i , employment L_i . Data-calibrated objects: factor shares α_i (labour), domestic bill coefficients $a_u = A_u^{bill}/\lambda_u$ with input-share matrix Ω (Ω_{iu} = the share of input u in sector i 's input bundle; rows sum to one), CPI weights ω_i ($\sum_i \omega_i = 1$), sector import margins m_i , saving rate s , exogenous government demand g^G , investment inv , exports X (no margin), the programme bundle g , and the labour bar $\bar{L} = \sum_i \bar{L}_i = 1$ (baseline income units, GDP = 1). Elasticities: intermediate substitution θ , labour–intermediate substitution ϵ , consumption substitution σ , allocation $\eta \in \{0, 1\}$ (ADR-0010), supply elasticity $\eta_s \geq 0$ (BETA only).

Price block. The CES intermediate price index and unit cost are

$$P_i(p) = \left(\sum_u \Omega_{iu} p_u^{1-\theta} \right)^{\frac{1}{1-\theta}}, \quad (\text{A1})$$

$$c_i(p, w; A) = \left(A_i^{\epsilon-1} (\alpha_i w^{1-\epsilon} + (1 - \alpha_i) P_i^{1-\epsilon}) \right)^{\frac{1}{1-\epsilon}}, \quad (\text{A2})$$

with the Cobb–Douglas limit $c_i = w^{\alpha_i} P_i^{1-\alpha_i} / A_i$ at $\epsilon = 1$. Zero profit requires

$$p_i = c_i(p, w; A) \quad \forall i. \quad (\text{A3})$$

At the calibration point $(p, w, A) = (\mathbf{1}, 1, \mathbf{1})$ the unit cost equals one exactly, because $\sum_u \Omega_{iu} = 1$ makes $P_i = 1$; this is the calibration identity used throughout.

Quantity block. With full cost-minimizing allocation ($\eta = 1$) the marginal-product condition inverts to

$$L_i = (p_i A_i^{(\epsilon-1)/\epsilon} \alpha_i^{1/\epsilon} y_i^{1/\epsilon} / w)^\epsilon = p_i^\epsilon A_i^{\epsilon-1} \alpha_i y_i / w^\epsilon, \quad (\text{A4})$$

At $\eta = 0$ the allocation is the frozen baseline, $L_i = \bar{L}_i$; under ADR-0020 option C the wage w in (A4) is then a per-sector w_i , solved from $L_i^{cm}(p, y_i, w_i) = \bar{L}_i$ jointly with (A3). Intermediate demand is charged with the *domestic* bill (ADR-0012):

$$int_i = p_i^{-\theta} \sum_u \Omega_{ui} p_u^\epsilon A_u^{\epsilon-1} P_u^{\theta-\epsilon} a_u y_u. \quad (\text{A5})$$

Household demand is CES with CPI deflation and the financing weights ds ($ds = 1$ except under F1, where $ds = d$, the preference tilt):

$$c_i = (1-s)(1-m_i)\omega_i ds_i E p_i^{-\sigma} / \sum_j \omega_j ds_j p_j^{1-\sigma}, \quad E = (1-\tau)w \sum_i L_i + F, \quad (\text{A6})$$

with the lump-sum tax rate $\tau = p \cdot g^G / (w \sum_i L_i)$ under F1/F3 and $\tau = p \cdot (g^G + g) / (w \sum_i L_i)$ under F2, and F the net external transfer entering expenditure after tax (ADR-0019). At $\eta = 0$ the wage bill is $\sum_i w_i \bar{L}_i$ in (A6). Saving sE leaks; the import content is charged margin by margin. Goods-market clearing reads

$$y_i = int_i + c_i + (1-m_i)(g_i^G + inv_i + g_i) + X_i, \quad (\text{A7})$$

where g_i appears only under the additive financings F2/F3 ($g_i = 0$ under F1; legacy manna is zero in the matrix).

The four regimes. The mobile systems solve for $X = (p, y, w, F) \in \mathbb{R}^{2N+2}$ with the numeraire $\Pi(p) = 1$, where

$$\Pi(p) = \left(\sum_i \omega_i p_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}}, \quad (\text{A8})$$

and close with the labour equation

$$\text{ALPHA: } \sum_i L_i(p, y, w) = \bar{L}, \quad (\text{A9})$$

$$\text{BETA: } \sum_i L_i(p, y, w) = \bar{L} \left[\frac{w/\Pi(p)}{w_0/\Pi_0} \right]^{\eta_s}, \quad w_0/\Pi_0 = 1. \quad (\text{A10})$$

The fixed-wage systems solve for $X = (p, y) \in \mathbb{R}^{2N}$ with the wage pinned $w \equiv 1$ and $F \equiv 0$; employment is a post-solve outcome. GAMMA runs them at the baseline elasticities, DELTA at the Leontief corner $(\theta_\delta, \epsilon_\delta, \sigma_\delta) = (10^{-4}, 10^{-4}, 10^{-4})$. BF is the sectoral-wage system at $\eta = 0$ (ADR-0020 option C, superseding the ADR-0019 $F = 0$ pin): the allocation stays frozen at the baseline, $L_i = \bar{L}_i$, and N sectoral wages w_i are solved from each sector's own first-order condition $L_i^{cm}(p, y_i, w_i) = \bar{L}_i$ jointly with zero profit, with unknowns $[p; y; w(1:N); F]$ and the external transfer F free. All rows in the matrix use $\eta = 1$ except BF.

A.3 The demand-only reduction

Lemma 1 (Price block demand-invariance). *The zero-profit system (A3) is homogeneous of degree one in (p, w) jointly and contains no demand variable. Consequently, for any technology A , every solution satisfies $p = w \pi(A)$, where $\pi(A) \in \mathbb{R}_{++}^N$ solves $c_i(\pi, 1; A) = \pi_i$ for all i ; relative prices and the ratio p/w are functions of A alone. Under the CPI numeraire, $w = 1/\Pi(\pi(A))$ and $p = \pi(A)/\Pi(\pi(A))$; under the fixed wage, $w = 1$ and $p = \pi(A)$. At $A \equiv \mathbf{1}$ the calibration identity gives $\pi = \mathbf{1}$, hence in both gauges*

$$p = \mathbf{1}, \quad w = 1, \quad \Pi = 1, \quad (\text{A11})$$

independently of the demand vector, of the financing closure, and of the shock size. The lemma is a statement about the single-wage price block: with a per-sector wage vector (the $\eta = 0$ endpoint, ADR-0020) homogeneity in (p, w) no longer holds sector by sector and the argument does not apply. The statement is scoped to the branch the numeraire selects: $\pi(A)$ denotes the normalised solution the kernel's solves return, and the equivalences below compare the systems on that branch (the CES price system at $\theta < 1$ with self-loops admits spiral branches in principle; no executed cell visits one).

Proof. Homogeneity: $c_i(tp, tw; A) = t c_i(p, w; A)$ for $t > 0$, by inspection of (A2)–(A1). Write $q = p/w$. Then (A3) is equivalent to $q_i = c_i(q, 1; A)$ for all i , i.e. $q = \pi(A)$, and the solution branch of (A3) selected by the numeraire is the ray $\{(w \pi(A), w)\}$. The CPI (A8) is homogeneous of degree one in p , so $\Pi(w \pi) = w \Pi(\pi)$; the numeraire $\Pi = 1$ or the pin $w = 1$ selects the scale. At $A \equiv \mathbf{1}$, $\pi = \mathbf{1}$ solves (A3) at $w = 1$ because $c_i(\mathbf{1}, 1; \mathbf{1}) = 1$ (the calibration identity), and $\Pi(\mathbf{1}) = 1$ because $\sum_i \omega_i = 1$. $\square$

Corollary 1 (Real-wage invariance). *At every solution of the shared block,*

$$\frac{w}{\Pi(p)} = \frac{1}{\Pi(\pi(A))}, \quad (\text{A12})$$

a function of technology alone. At $A \equiv \mathbf{1}$ this equals $1 = w_0/\Pi_0$: the equilibrium real wage sits at its anchor for every demand vector, every financing closure and every shock magnitude. (Measured in the matrix: the GDP deflator reads 1.000000 in the twelve $\eta = 1$ cells, and a demand shock of ten times the programme size moves $\max |p - \mathbf{1}|$ by 2.7×10^{-15} ; docs/ETAs.md.)

Lemma 2 (Elasticity-free affine quantity block). *Fix $\eta = 1$ and $A \equiv \mathbf{1}$, and evaluate the quantity*

block at the demand-only prices $(p, w) = (\mathbf{1}, 1)$. Then for every value of $(\theta, \epsilon, \sigma)$:

$$\text{int} = \Omega^\top \text{diag}(a) y, \quad L = \text{diag}(\alpha) y, \quad \sum_i L_i = \alpha^\top y, \quad (\text{A13})$$

$$c = (1 - s) \text{diag}(1 - m) \tilde{\omega}(ds) E, \quad E = \alpha^\top y - \sum g^G + F \quad (F1/F3),$$

where $\tilde{\omega}(ds) = (\omega \odot ds) / \mathbf{1}^\top (\omega \odot ds)$ and $E = \alpha^\top y - \sum g^G - \sum g$ under F2. Hence the clearing system (A7) is exactly affine,

$$y = G y + b(F), \quad G = \Omega^\top \text{diag}(a) + (1 - s) \text{diag}(1 - m) \tilde{\omega}(ds) \alpha^\top, \quad (\text{A14})$$

$$b(F) = (1 - m) \odot (g^G + \text{inv} + g) + X + (1 - s) \text{diag}(1 - m) \tilde{\omega}(ds) (F - \sum g^G - \mathbf{1}_{F2} \sum g), \quad (\text{A15})$$

with g present only under the additive financings F2/F3 and $\mathbf{1}_{F2}$ selecting the tax-financed column (the bracket is $F - \sum g^G$ under F1/F3 and $F - \sum g^G - \sum g$ under F2). No elasticity parameter enters G or b . At $\eta = 0$ the aggregate is still pinned, $\sum_i L_i = \bar{L}$, but the unit-price reduction below describes only the retired common-wage pin: there E did not depend on y and the household round dropped out, $G_{BF} = \Omega^\top \text{diag}(a)$. Under the ADR-0020 sectoral-wage closure the wages w_i move with the demand composition, the prices leave the unit point, and this affine form does not describe the $\eta = 0$ row.

Proof. At $p = w = A = \mathbf{1}$: $P_u = 1$ in (A5), so all elasticity factors p^ϵ , $P^{\theta-\epsilon}$, $p^{-\theta}$ equal 1 and (A5) reduces to the fixed-coefficient round $\sum_u \Omega_{ui} a_u y_u$; (A4) reduces to $L_i = \alpha_i y_i$ since every exponent collapses at unit prices; in (A6) the terms $p_i^{-\sigma}$ and the normalizer reduce to 1 and $\sum_j \omega_j ds_j$, giving the fixed spending weights $\tilde{\omega}(ds)$; and the tax rate $\tau = \sum g^G / \alpha^\top y$ turns E into the affine form shown. Collecting terms in (A7) gives (A14)–(A15): the intermediate round contributes the columns $\sum_i \Omega_{ui} a_u = a_u$ (row sums of Ω equal 1) and the household round contributes $(1 - s) \sum_i (1 - m_i) \tilde{\omega}_i(ds) \alpha_u$ per unit of y_u . $\square$

Remark 1 (Nonsubstitution). *Lemma 1 and Lemma 2 are the production side of the Samuelson nonsubstitution theorem: with one primary factor, constant returns and no joint production, the cost-minimizing technique — and hence relative prices — is independent of the demand vector, and final demand allocates quantities only (Samuelson 1951). The kernel’s nested-CES structure is a smooth parametrisation of that corner; the demand-only matrix therefore inherits the theorem’s rigidity, and this is what docs/DOCS_ASSESSMENT.md section 4.2 records as the “model-class property”*

behind the coincidences. The IO endpoint makes the rigidity literal — the Robinson (2006) fixed-price multiplier — but, as Proposition 2 shows, it is not an approximation: the same candidate equilibrium $(p, y) = (\mathbf{1}, y^*)$ solves the fixed-wage system at every elasticity triple in the fixed-wage gauge.

A.4 The two equivalences

Proposition 1 (ALPHA $\equiv$ BETA). *Let $A \equiv \mathbf{1}$ and $\eta = 1$. The ALPHA system (A3),(A7),(A9), $\Pi = 1$ and the BETA system (A3),(A7),(A10), $\Pi = 1$ have identical solution sets, for every $\eta_s \geq 0$ and every financing closure.*

Proof. The two systems differ in exactly one equation. By Lemma 1 and Corollary 1, every solution of the shared block satisfies $w/\Pi = 1 = w_0/\Pi_0$, so BETA’s labour equation (A10) reduces to $\sum_i L_i = \bar{L}$, which is (A9). Conversely, any solution of the ALPHA system satisfies BETA’s equation by the same substitution. The systems are therefore equivalent, equation by equation. $\square$

Proposition 2 (GAMMA $\equiv$ DELTA). *Let $A \equiv \mathbf{1}$, $\eta = 1$, $F \equiv 0$, $w \equiv 1$. Then $(p, y) = (\mathbf{1}, y^*)$ with $y^* = (I - G)^{-1}b(0)$ solves the fixed-wage system at every elasticity triple $(\theta, \epsilon, \sigma)$; the employment outcome is $L^* = \alpha^\top y^*$, the consumption block is $c^* = (1-s) \text{diag}(1-m) \tilde{\omega}(ds) (L^* - \sum g^G - \sum g \mathbf{1}_{F2})$, where $\mathbf{1}_{F2}$ selects the tax-financed column, and the block decomposition (A13) holds verbatim. In particular GAMMA (baseline elasticities) and DELTA (the Leontief corner) share this solution, and the analytic system (A14)–(A15) — `leontief_multiplier` in `src/closures/labor/labor.jl` — computes it exactly.*

Proof. Existence and form: at $p = \mathbf{1}$ the price block holds by the calibration identity (Lemma 1), and by Lemma 2 the clearing block at $p = \mathbf{1}$ is the affine system, of which $y^* = (I - G)^{-1}b(0)$ is the solution. Uniqueness on the admitted branch: the scale-determinacy guard (ADR-0017) verifies at the reference prices that $(I - G)$ is nonsingular — contraction if $\max_u \text{colsum}_u(G) < 1$, otherwise a direct rank test — and that the candidate y^* is strictly positive; the guard admits the branch on which the affine fixed point is unique and positive, but does not by itself exclude further roots of the nonlinear system; the multi-init admissibility checks recorded for the F1 column (three distinct starting points converging to the same root) are the evidence that the reported root is the admitted one. Both the GAMMA and the DELTA solve are subject to the same guard, and both converged to this root (residuals 4.4×10^{-16} to 2.0×10^{-13} across the six fixed-wage cells, `runs/matrix_5x3-v10-{\GAMMA,DELTA}-*/manifest.toml`). The elasticity-independence is exact

in the equations, not an ϵ -limit artefact: the DELTA solve at $\epsilon_\delta = 10^{-4}$ reproduces the analytic y^* with relative error 0.0 (F2/F3 verification, recorded in `docs/DOCS_ASSESSMENT.md` section 4.2), and the v10 flow metrics of the two rows agree to all printed decimals with residual-level wedge differences $\sim 10^{-12}$. $\square$

Corollary 2 (Classification). *For the manuscript’s framing of the matrix:*

<i>Coincidence</i>	<i>Status</i>	<i>What is unidentified</i>
$GAMMA \equiv DELTA$	definitional in form, substantive in mechanism (Lemma 2)	$(\theta, \epsilon, \sigma)$ in the fixed-wage gauge
$ALPHA \equiv BETA$	derivable (Cor. 1)	η_s in the flexible-wage gauge
$F2 \equiv F3$ (mobile rows)	derivable (Section A.6)	the financing label; $F_{F3} = F_{F2} - B_{gov}$
BF vs $ALPHA$	not a coincidence: ADR-0020 option C gives the frozen allocation a price channel, so it is a distinct economy	— (the $\eta = 0$ row is an identified equilibrium: own prices, wage vector 0.97–1.67, own position)
sectoral corner $\equiv BF$ ($\eta_{s,i} = 0$; Prop. 3)	exact, by construction (Prop. 3)	— (the corner is the endpoint: the same equations)

The first three entries are not numerical accidents; they hold at machine precision in the executed generation, and none is a mis-specification. They are properties of the experiment: a demand-only programme in a one-factor CRS economy cannot identify any elasticity of the kernel (the mechanism is Remark 1). The fourth entry is different in kind: the $BF/ALPHA$ coincidence held only in the superseded $F = 0$ pin generation (ADR-0019 D2), where the frozen allocation had no channel to prices; under ADR-0020 option C the $\eta = 0$ row carries sectoral wages and is an identified equilibrium, so no BF quantity is left unidentified and none must be withheld. The fifth entry is an equivalence between two closures rather than between two rows of the executed matrix, and it is exact by construction: the sectoral family’s rigid corner reproduces the $\eta = 0$ endpoint, which is why no separate rigid sectoral cell is reported.

Remark 2 (Off the demand-only slice). *The equivalences fail the moment technology moves. At*

$A \neq \mathbf{1}$, Corollary 1 gives $w/\Pi = 1/\Pi(\pi(A)) \neq 1$ in general, and BETA's labour equation reduces to $\alpha^\top y = \bar{L} [1/\Pi(\pi(A))]^{\eta_s}$: employment is pinned to a technology-determined constant that now depends on η_s . The supply-side pilot (+20% in sector 1) measures exactly this separation — the failure of Proposition 1's hypothesis $A \equiv \mathbf{1}$: $w = 1.004903$, with $L = 1.0024$ at $\eta_s = 0.5$ versus $L = 1.0098$ at $\eta_s = 2$ against ALPHA's $L = 1.0000$ (*docs/ETAs.md*). The identification of η_s therefore requires a supply-side scenario — the premise of the supply arm.

A.5 The sectoral family and its rigid corner

The two equivalences above are coincidences *within* the executed matrix. This section records a third, of a different kind: an equivalence between two *closures*, one of which is the matrix's $\eta = 0$ endpoint and the other a corner of a family that extends beyond the matrix. It is exact rather than asymptotic, and it is what makes the labour door of Section A.7 concrete.

The family. Replace the single aggregate supply equation (A10) by N sectoral labour markets,

$$L_i^{cm}(p, y_i, w_i) = \bar{L}_i \left[\frac{w_i}{\Pi(p)} \right]^{\eta_{s,i}}, \quad i = 1, \dots, N, \quad \eta_{s,i} \geq 0, \quad (\text{A16})$$

solved jointly with zero profit (A3), clearing (A7), the numeraire $\Pi(p) = 1$ and a free transfer F , in the unknowns $X = (p, y, w(1:N), F) \in \mathbb{R}^{3N+1}$ (ADR-0022). The scalar closure (A10) is the member of this family with one common wage and $\eta_{s,i} \equiv \eta_s$; the matrix's BETA row is therefore the scalar *special case* of a family, not a separate mechanism. Because the N wages enter the N zero-profit conditions (A3) through the price index (A1), the price block now depends on the N employment levels and Lemma 1 no longer applies: demand reaches prices through the wage structure, $p = \pi(A, w)$.

Two rigidities, one equation. In both closures labour is immobile, but for different reasons. At the $\eta = 0$ endpoint the allocation is frozen *by assumption*: $L_i = \bar{L}_i$ is a datum of the quantity block, and the sectoral wage is whatever clears cost-minimizing demand against it. At $\eta_{s,i} = 0$ each sector has a *vertical* supply curve, $L_i^s = \bar{L}_i$ for every w_i , so the market clears at the same allocation with a demand-determined wage. A constraint imposed on a quantity and an infinitely inelastic price response turn out to be the same row of algebra — which is the content of the proposition below.

Proposition 3 (The rigid corner is the $\eta = 0$ endpoint). *Set $\eta_{s,i} = 0$ for every i . Then the sectoral system (A16) and the $\eta = 0$ endpoint of ADR-0020 option C have the same solution set.*

Proof. Write both systems in the unknowns $X = (p, y, w, F)$ as four row blocks:

$$(P) \quad p_i = c_i(p, w_i; A), \quad i = 1, \dots, N, \quad (A3)$$

$$(Y) \quad y_i = int_i + c_i + (1 - m_i)(g_i^G + inv_i + g_i) + X_i, \quad i = 1, \dots, N, \quad (A7)$$

$$(W) \quad \log L_i^{cm}(p, y_i, w_i) - \log \bar{L}_i - \eta_{s,i} \log \frac{w_i}{\Pi(p)} = 0, \quad i = 1, \dots, N,$$

$$(E) \quad \Pi(p) = 1.$$

The quantity block (Y) draws labour income from the wage bill $\sum_i w_i L_i$ inside (A6) and from the tax rate τ ; this is the only place, besides (W), where the two systems differ. The $\eta = 0$ endpoint is the same four blocks with two changes: (W) drops the supply term ($\eta_{s,i} = 0$), and (Y) evaluates the wage bill at the *frozen* allocation, $L_i := \bar{L}_i$, instead of at the cost-minimizing demand L_i^{cm} .

(i) Let X solve the sectoral system at $\eta_{s,i} \equiv 0$. Its (W) gives $L_i^{cm}(X) = \bar{L}_i$ for every i , the supply term vanishing identically. Substituting $L_i^{cm} = \bar{L}_i$ into its (Y) makes that block equal the endpoint's (Y); (P) and (E) are identical in the two systems by inspection. Hence X solves the endpoint system.

(ii) Let X solve the endpoint system. Its (W) gives $L_i^{cm}(X) = \bar{L}_i$, which satisfies the sectoral (W) at $\eta_{s,i} \equiv 0$; substituting it into (Y) yields the sectoral (Y). Hence X solves the sectoral system. Both systems are square — $3N + 1$ unknowns, $3N + 1$ equations — and their residual maps agree on the common solution set, so the solution sets coincide. $\square$

Remark 3 (Why the family interpolates, and where it does not close). *Sweeping $\eta_{s,i}$ moves the adjustment from quantities to prices. At $\eta_{s,i} = 0$ the allocation is pinned and the N wages absorb the shock; as $\eta_{s,i} \rightarrow \infty$ the supply curves become infinitely elastic, the real wage returns to its technology-determined anchor and employment becomes the absorbing margin — GAMMA-like prices and GAMMA's absorbing margin. The two ends of the family are therefore the matrix's own rigidity rows: BF (allocation frozen, wages free) at the rigid corner, GAMMA (real wage pinned, allocation free) at the elastic one. Two qualifications are exact and matter for reading the table below. First, the elastic end is a limit, not a cell: the supply slope diverges and the direct solve eventually fails ($\eta_s = 10^6$; probe13). Second, the family keeps the transfer F free, whereas the fixed-wage system pins $F \equiv 0$; the family therefore spans the two rigidity rows without terminating in either.*

Measured. Table A1 reports the executed sectoral cells (matrix_5x3-v10-BETA-*; ADR-0022). Four readings.

(a) *Prices move, and with the demand composition.* Every sectoral cell carries a deflator different from one and a $\max |p-1|$ that differs between F1 and F2/F3 — the signature no single-wage closure produces (all twelve $\eta = 1$ cells of the matrix read exactly 0.000000). (b) *The elasticity splits the adjustment.* As η_s rises from 0.25 to 2, the price response falls monotonically (0.1529 $\rightarrow$ 0.0372 at F2) while employment rises (1.000864 $\rightarrow$ 1.001914): the family trades price adjustment for quantity adjustment, and the welfare cost falls with it (−1.69% $\rightarrow$ −1.39%). The F1 column, whose programme is financed by a preference tilt rather than a tax, is the exception: its welfare index is positive (+0.42% at $\eta_s = 0.5$). (c) *Incidence matters more than the level.* Making the *programme* sectors rigid ($\eta_{s,i} = 0$ there, $\eta_s = 0.5$ elsewhere) more than doubles the price response (0.2720 against 0.1057) and *reverses* the employment effect: employment falls by 0.34% and the welfare index by 2.69%, against +0.13% and −1.57% for the uniform cell. Rigidity on the largest half by baseline employment instead gives 0.1234 with a near-neutral +0.02%. (d) *Wage dispersion is large.* The mechanism moves relative wages: $\max_i w_i / \min_i w_i$ reads 1.26 at $\eta_s = 0.5$ and 1.70 for the rigid-programme variant, against 1.72 at the $\eta = 0$ endpoint — for a programme worth 1.33% of GDP.

[Table A1 about here.]

The corner is the endpoint, numerically. Solving the sectoral system at $\eta_{s,i} = 0$ from the $\eta = 0$ solution reproduces (p, y, w, F) exactly — $\max |\Delta| = 0.00 \times 10^0$ in all four blocks, and below 10^{-10} from a cold start (`tests/test_sectoral_labour.jl`; ADR-0022) — as Proposition 3 requires. No separate rigid sectoral cell is therefore reported: the corner *is* the BF row of Table A2.

The proof never uses $A = 1$. Nothing in the four row blocks of the proposition’s proof refers to the technology shock, so the nesting is not a demand-only property. Verified directly: under a +20% productivity shock in sector 1 the canary again reads 0.00×10^0 in all four blocks (`probe15_sectoral_supply.jl`; the figures are recorded in `docs/log/2026-09.md` and `docs/ETAs.md`).

The same probe shows why $\eta_{s,i}$ matters outside the demand-only slice: the rungs separate under the shock (employment 1.000000, 1.002187, 1.008218 and the real-wage spread 1.215, 1.105, 1.043 at $\eta_s = 0, 0.5, 2$), so the sectoral elasticity is identifiable only with a supply-side scenario, while under a demand-only programme it is a sensitivity parameter.

A.6 What actually differs across the rows

Lemma 2 shows that the demand-only matrix is a partition of *scalar closure regimes* over one and the same linear system. Two scalar regimes occur across the $\eta = 1$ rows, and the $\eta = 0$ endpoint is now a third, genuinely nonlinear regime (the sectoral-wage system):

Rows	F	Employment	Household round in G
ALPHA, BETA ($\eta = 1$)	free (instrument)	pinned, $L = \bar{L}$	active
GAMMA, DELTA ($\eta = 1$)	$F \equiv 0$	endogenous, $L = \alpha^\top y$	active
BF ($\eta = 0$)	free (solved)	pinned via allocation	active (wage bill)

The F -instrument. In the mobile regime the labour pin selects the transfer. Writing $y(F) = (I - G)^{-1}(b_0 + fF)$ with $f = (1 - s) \text{diag}(1 - m) \tilde{\omega}(ds)$, the pin $\alpha^\top y = \bar{L}$ gives

$$F^* = \frac{\bar{L} - \alpha^\top (I - G)^{-1} b_0}{\alpha^\top (I - G)^{-1} f}. \quad (\text{A17})$$

This instrument is why the mobile F2 and F3 columns are real-neutral: F2 charges the programme through the lump-sum tax ($E = \alpha^\top y - \sum g^G - \sum g$), F3 through the external account ($E = \alpha^\top y - \sum g^G + F$), and the two b -vectors differ by exactly $f \sum g$. The pin (A17) therefore yields $F_{F3}^* = F_{F2}^* - \sum g$, identical real allocations, and an identical net external position $F + B_{gov}$. Measured: $F_{F3} - F_{F2} = -0.01330991 = -G_0$ to eight decimals in every mobile row (the F values are printed in `paper/tables/matrix_5x3_v6_flows.md`, Table 2, which the v10 re-mint reproduces to all printed decimals; $G_0 = \sum_i g_i = 0.013310$, ADR-0024).

Who pays. The economically meaningful difference between the two paradigms is which margin absorbs the programme. In the mobile rows the labour pin forces the household to pay (consumption -1.82% under F2/F3; the F instrument crowds it out exactly), while real GDP is flat by construction ($w \bar{L}$ deflated, ADR-0018). In the fixed-wage rows employment is the margin: F3 expands employment by 1.78% and consumption by 2.26% (the Keynesian case), F2 leaves GDP nearly neutral (+0.13%) with a welfare loss from the tax, and F1 shifts composition slightly downward (-0.10% employment). The $\eta = 0$ row is no longer degenerate: at the sectoral-wage endpoint the welfare index reads +0.096% (F1) and -1.980% (F2/F3), the deflator moves (1.0063 / 1.0072), and the external account closes in every column. Two caveats from the ADR-0020 audit now describe the *superseded* v5 pin generation: there the recorded BF/ALPHA differences were carried by the pin, not by the frozen allocation (replacing the pin by the labour equation reproduced the ALPHA

cells to 2.8×10^{-17}), and the $\eta = 0$ external position was not identified (every pin value $F = c$ an exact root that moved the booked position one-for-one). Option C resolves both: the account closes to $\leq 1.04 \times 10^{-11}$ at BF, F is a solved equilibrium object, financing neutrality extends to $\eta = 0$ ($F_{F3} = F_{F2} - B_{gov}$ to 3.5×10^{-14}), and no BF entry need be withheld.

Admissibility. All fifteen cells execute (v6 and, re-minted, v10). In particular the F1 column is admissible in the fixed-wage rows: the saving rate and the import margins supply the marginal leakage that keeps $\max \text{colsum}(G) < 1$ (ADR-0017's actual-matrix criterion reads 0.8314 on full-71 and 0.8320 under the F1 tilt); the endogenous tax rate $\tau = \sum g^G / \alpha^\top y$ is a fixed-revenue level ($T = p \cdot g^G$) and enters b , not G (the retired heuristic guard had blocked these cells in the v1 generation).

A.7 Scope conditions: what breaks the equivalences

Both propositions rest on Lemma 1: the price block contains no demand variable. Any model change that lets demand into that block restores identification and creates variety in the published matrix. The doors are:

1. **Labour door — sector-specific wages.** With N sectoral labour markets $w_i = \phi_i(L_i)$ instead of one common w , the homogeneity argument of Lemma 1 collapses: $L_i = \alpha_i y_i$ depends on demand, so w_i and through zero profit p_i become demand-sensitive. Zero profit is preserved (no rents), the new parameters are sectoral supply elasticities of the same family as η_s , and the variant repairs the BF row's common-wage defect (the `labor.BF` open gate of `registry/closures.toml`, resolved by ADR-0020 option C; `cbase2/review.md` section 1). Ranked candidate 1 in `docs/ETAs.md`. ADR-0020 option C is this door, adopted at the $\eta = 0$ endpoint: it is why the v6 BF row is a distinct economy rather than a normalisation of ALPHA. ADR-0021 (proposed) reuses the vector-wage apparatus in the fixed-wage row, but with an *exogenous* pin — that delivers design heterogeneity, not demand-sensitivity, because the pinned vector keeps the price block demand-free. ADR-0022 (accepted) generalises this door across the $\eta = 1$ rows: N sectoral labour markets with supply elasticities $\eta_{s,i}$ (Section A.5), whose rigid corner is precisely the option C endpoint proved in Proposition 3. It breaks the single-wage homogeneity on which Lemma 1 rests, requires no income closure (no rents are created), and its scalar member is the existing BETA row.
2. **Markup door — demand-sensitive markups.** A pricing rule $p_i = \mu_i(y_i) c_i(p, w)$, e.g.

$\mu_i = 1 + \kappa(y_i/y_{i0} - 1)$, breaks Lemma 1 directly. The real cost is not code but accounting: markup income is profit, so a profit-income-and-spending closure is required or the external-account identity behind the canary breaks. Nested at $\kappa = 0$. Candidate 2 in `docs/ETAs.md`.

3. **Capacity door — a fixed factor, or its reduced form.** A sector-fixed factor with a rental schedule makes marginal cost rise in y_i (rents require a rental-income closure — the largest surgery, candidate 3 in `docs/ETAs.md`). The reduced-form variant avoids all income-closure problems: an endogenous utilization externality $A_i^{eff} = A_i(y_i/\lambda_i)^{-\delta}$ raises unit cost with demand while keeping zero profit (the extra cost is paid to no one), so a one-line change in the cost hook (A2) suffices. With the opposite sign ($\delta < 0$) it is the Kaldor–Verdoorn case: increasing returns make prices *fall* with demand — heterodox-friendly variety in the other direction. (New candidate; see Section A.8.)

4. **External door — endogenous import prices.** An upward-sloping foreign supply curve $p_i^M = p_{i0}^M (M_i/M_{i0})^\zeta$, $\zeta \geq 0$, enters the intermediate index (A1) and the CPI through the margins, so π becomes a function of import volumes and hence of demand. The surgery is moderate and it exploits precisely the external-account machinery the paper has just built (ADR-0019): the programme’s import content would bid up import prices — a structuralist absorption/term-of-trade channel that fits the open-economy framing. (New candidate.)

5. **Technology door — supply shocks.** $A \neq 1$ breaks Corollary 1 and separates BETA from ALPHA (Remark 2); this is the supply arm already designed in `docs/ETAs.md`, not a repair of the demand-only matrix.

Corollary 3 (No-go for nominal wage rules). *Within the kernel’s one-factor CRS price block, no rule that writes the nominal wage as a function of endogenous aggregates — an aggregate wage curve $w = \bar{w}(\alpha^\top y/\bar{L})^\varphi$, nominal indexation, or a re-anchored numeraire — can create demand-sensitive real outcomes. By Lemma 1, $p = w\pi(A)$ and $\Pi(p) = w\Pi(\pi)$ at every solution, so the real wage $w/\Pi = 1/\Pi(\pi(A))$ is invariant to whatever sets w ; the rule merely re-labels the nominal scale. With the CPI numeraire or the $w = 1$ pin such a rule is either vacuous or over-determines the system (in the fixed gauge it forces $\alpha^\top y = \bar{L}$, i.e. silently reverts GAMMA to ALPHA). A rule on the real wage, $w/\Pi = \phi(L)$, is BETA’s equation (A10) rearranged and is already in the model — where it is pinned by technology under demand-only shocks. The numeraire lever is likewise dead (DE-0011). A single common wage curve is also what a single-wage price block can express; the sectoral-wage door above escapes the corollary precisely by breaking the single-wage homogeneity on which it rests.*

Quantity-side levers (endogenous saving rates, endogenous export demand) change G or b but not the price block; they add variety between financing columns, not between closures, and leave Lemma 1 intact.

A.8 Numerical evidence

The fifteen matrix cells were re-executed as the `matrix_5x3_v9` generation on the extended kernel that carries the sectoral family (ADR-0022), and are cited here from the re-minted `matrix_5x3_v10` generation (the C1-corrected measurement, 2026-09-20). `v6` and `v9` are bit-identical to each other ($\max|\Delta| = 0$ on `gdp_rel`, `consumption_rel`, employment, the deflator, $\max|p - 1|$, the external transfer, the wage extremes and the identity residual); `v10` reproduces them to solver precision ($\max 9.34 \times 10^{-9}$ at the stiff BF-F1 `wage_max` diagnostic, 1.63×10^{-11} in the worst ALPHA/BETA/GAMMA/DELTA row, 4.55×10^{-15} at the exact-baseline cells; no printed figure is affected, `paper/tables/matrix_5x3_v10_flows.md`, Table 2), so the equivalences and the table below are invariant to the extension. The sectoral cells of the same generation are reported separately in Table A1.

Table A2 extracts the rows of the executed `matrix_5x3_v10` generation (the fifteen matrix cells reproduce their `v6/v9` manifests to all printed decimals) (income-side real GDP and the household-consumption welfare index, ADR-0018; employment). The paired rows (ALPHA/BETA, GAMMA/DELTA) are identical to all printed decimals in every reported metric, including the full flow decomposition (saving, tax, investment plus exports, imports, taxes on intermediates; see `paper/tables/matrix_5x3_v6_flows.md`, whose matrix rows the `v9` and `v10` generations reproduce to all printed decimals). The three BF rows are included for contrast: the $\eta = 0$ sectoral-wage endpoint is a distinct economy, not a normalisation of ALPHA. The GDP deflator reads 1.000000 in the twelve $\eta = 1$ cells — the numerical face of Lemma 1 — while the three BF $\eta = 0$ cells carry a deflator of 1.006326 (F1) and 1.007181 (F2/F3), the price response the sectoral wages create.

[Table A2 about here.]

Implications for the manuscript. (i) The two-paradigm reading is safe to state as a theorem-backed claim: the flexible-wage rows form one equivalence class (composition at pinned employment), the fixed-wage rows another (employment at pinned real wage), and the difference between the classes is the F -instrument — a derivation, not an observation. (ii) The null results for η_s and for $(\theta, \epsilon, \sigma)$ under demand-only shocks should be reported as identification statements (Corollary 2), which is

precisely what reviewer 2’s critique needs. (iii) The variety program for the demand-only matrix must route through one of the five doors of Section A.7; within it, the capacity door’s utilization/Verdoorn variant and the external door are the cheap additions beyond the three candidates already ranked in `docs/ETAs.md`, while Corollary 3 rules out the entire nominal-wage-rule family *ex ante*. The labour door is no longer hypothetical: ADR-0020 option C delivers it at the $\eta = 0$ endpoint, and the general sectoral-labour-market form extends it across the $\eta = 1$ rows. (iv) The labour door is now an implemented mechanism rather than a proposal: the sectoral family of Section A.5 is promoted, tested and executed (ADR-0022), it delivers demand-sensitive prices inside a flexible-wage closure, and it leaves the executed matrix unchanged to solver precision (no printed figure affected). The manuscript can therefore state three exact equivalences and present the family as the answer to the variety question — with two caveats that must travel with it: under demand-only shocks $\eta_{s,i}$ is a *sensitivity* parameter, not an estimate, so the level of the price response is identified only by the supply arm (Remark 2); and the rule that makes a subset of sectors rigid is a modelling choice, to be reported as a band rather than a point.

B Kernel, calibration, and A-bill accounting

Reproducibility

C Matrix variants and gamma/beta representation robustness

D Vector construction and data provenance

E Closures and the neo-Keynesian mapping

This appendix maps the CGE/IO cleavage and the neo-Keynesian alternative — the EU JRC FIDELIO model — onto the closure matrix. It supports Section 3.

E.1 The core theoretical cleavage: demand shocks in CGE versus IO models

A central question in applied input–output modeling — especially for evaluating green public investment programmes such as the EU Recovery and Resilience Facility or large-scale building renovation

— is whether an exogenous demand stimulus:

1. **expands real aggregate output** by mobilising unutilised resources (the Keynesian / Leontief perspective), or
2. **merely reallocates scarce resources**, causing factor prices to rise and crowding out private activity with virtually zero net change in aggregate real GDP (the canonical neoclassical CGE perspective).

Why standard CGE models produce minimal aggregate effects from demand shocks. In textbook Walrasian CGE models: (i) *full factor utilisation* — aggregate primary factor endowments $(\bar{L}, \bar{K})$ are exogenously given and fully employed; (ii) *instantaneous market clearing* — prices and wages are perfectly flexible and clear all commodity and factor markets instantaneously; (iii) *Say’s Law operates* — output is supply-determined by the production possibility frontier $Y = F(K, L, \mathbf{X})$; (iv) *crowding out* — an exogenous increase in public investment raises demand in targeted sectors, but because total labour and capital cannot expand, the extra factor demand bids up wages and capital rentals, and the relative price increase squeezes the margins of non-shocked sectors, crowding out private consumption, private investment or net exports. The aggregate real GDP response is negligible ($\Delta Y \approx 0$).

E.2 The macroeconometric alternative: the EU JRC FIDELIO model

The **FIDELIO** model (*Fully Interregional Dynamic Econometric Long-term Input–Output*; JRC / DG GROW; Rocchi et al., 2019, 2025) was built specifically to overcome this limitation. Rather than adopting a Walrasian general-equilibrium core, FIDELIO is a *dynamic macroeconomic, multi-sectoral Neo-Keynesian* model grounded in the official Eurostat/JRC FIGARO inter-country input–output tables (64 NACE industries, 46 countries/regions). It generates positive, non-trivial real output and employment multipliers from demand shocks through five structural features:

1. **Involuntary unemployment and the wage-setting curve.** Wages do not clear the labour market. The baseline economy features involuntary unemployment ($u > 0$). Nominal wages adjust sluggishly along an econometrically estimated *wage curve* (Blanchflower–Oswald tradition):

$$\Delta \ln w_{i,r,t} = f(\text{inflation}_{t-1}, \text{productivity}_t, u_{r,t-1}).$$

Because labour supply is not vertical at baseline, firms can hire idle workers to expand pro-

duction without hitting an immediate supply wall.

2. **Mark-up pricing and sluggish cost pass-through.** Product prices are set via markups over unit costs (derived from flexible Translog cost functions over capital, labour, energy and intermediate inputs) rather than jumping to Walrasian clearing levels. Output in the short-to-medium run is *demand-determined*.
3. **Keynesian multiplier loops (the ECM consumption block.)** Household consumption follows a dynamic error-correction model tied to current real disposable income rather than to an intertemporal Euler equation:

$$\Delta G \Rightarrow \uparrow Y \Rightarrow \uparrow L \Rightarrow \uparrow (w \cdot L) \Rightarrow \uparrow C^{\text{household}},$$

inducing a secondary Keynesian expenditure loop (a Type-II multiplier), yielding aggregate output multipliers between 1.2 and 1.8 in JRC policy simulations.

4. **Dynamic capital accumulation (endogenous capacity).** Capital is not a fixed static endowment; investment responds dynamically through an accelerator / user-cost formulation, so sustained demand increases capacity utilisation and expected returns, prompting firms to invest and expand capital stocks over time.
5. **Medium-to-long-run re-equilibration.** As unemployment drops and capacity tightens, the wage curve drives up wages and unit costs; the resulting rise in output prices erodes export competitiveness, generating gradual crowding out toward a medium-run NAIRU — unless the stimulus was in public capital or R&D, in which case it permanently shifts potential output outward.

E.3 Mapping into the BeyondHulten 5×3 closure matrix

The platform organises macroeconomic closures across two dimensions (five labour closures $\times$ three financing closures).

The five labour closures.

1. **ALPHA** (full-employment mobile CGE benchmark): $\sum_i L_i = \bar{L}$, single flexible wage w , cost-minimising intersectoral allocation ($\eta = 1$).
2. **BF** (immobile-labour endpoint selector): $\eta = 0$ freezes sectoral labour quantities to baseline

shares $L_i = \bar{L}_i$; with the sectoral-wage endpoint (ADR-0020) the wage vector is free and the frozen allocation carries a price channel.

3. **BETA** (elastic labour supply on the real wage): $\sum_i L_i = \bar{L} [(w/P)/(w_0/P_0)]^{\eta_s}$, deflated by the CPI.
4. **GAMMA** (fixed real wage, uncapped extensive margin): $w/P = \bar{w} = 1$, employment $L = \sum_i L_i$ completely demand-determined.
5. **DELTA** (Leontief IO corner): GAMMA ($w/P = 1$) combined with the zero-substitution limit of the CES production core ($\theta, \epsilon, \sigma \rightarrow 0^+$), reproducing the fixed-price Leontief multiplier.

(Complementarity extension **ZETA**: $0 \leq \bar{L} - L \perp w/P - \bar{w} \geq 0$, regime-switching between GAMMA slack and ALPHA full employment; recorded as an idea, not a closure.)

The three financing closures.

1. **F1** (preference reallocation / demand tilt): budget-neutral composition shift within fixed household expenditure ($\sum_i p_i c_i = E_h$); no net fiscal injection.
2. **F2** (tax-financed public investment): balanced government budget ($\sum_i p_i g_i = T(p)$); autonomous public investment financed by an explicit domestic tax counterparty.
3. **F3** (external / debt-financed programme): autonomous public spending financed externally or via debt ($\sum_i p_i g_i = F$), with imports and external balances absorbing the deficit.

Matrix location of models.

[Table A3 about here.]

- **Where standard CGE sits:** ALPHA-F1 or ALPHA-F2 — fixed aggregate labour $\bar{L}$ and flexible wages force crowding out; real output effects are close to zero.
- **Where FIDELIO sits:** in its static / basic IO mode (modules 1–2) at DELTA-F3 (a pure Leontief input–output multiplier driven by external funding); in its short-run dynamic mode at GAMMA-F3 (fixed real wage, unconstrained extensive margin, autonomous injection), activating the full Keynesian responding multiplier; and, in the medium-to-long run, moving dynamically from GAMMA-F3 toward BETA-F3 as the wage curve drives real wage growth.

E.4 How neo-Keynesian assumptions map into our closures

In this platform, neo-Keynesian assumptions are not auxiliary parameters — they define the mathematical structure of the closure.

A. Wage rigidity (extensive margin versus wage inflation). In accordance with ADR-0014 and DE-0004, wage rigidity here is defined on the *real wage* ($w/P = \bar{w}$), deflated by the CPI numeraire ($P = 1$). In GAMMA, fixing $w/P = 1$ removes the wage as an equilibrating price: the labour-market equation changes from clearing $L = \bar{L}$ to an unconstrained demand equation $L = \sum_i L_i(y_i, \mathbf{p})$, and the extensive margin of employment absorbs the demand shock. Asymmetric rigidity (ZETA) follows the spirit of Baqaee–Farhi (2022): downward wage rigidity creates a regime switch — demand deficits cause unemployment (GAMMA regime), while demand surges that exhaust slack trigger wage inflation (ALPHA regime).

B. Price rigidity and fixed markups. In standard New-Keynesian models, Calvo or Rotemberg pricing prevents prices from adjusting instantaneously to demand shocks. Here, under GAMMA, wages are pegged ($w = 1$) and the shock is demand-driven, so marginal costs remain roughly constant and product prices exhibit high inertia ($\mathbf{p} \approx 1$). In DELTA, the zero-substitution limit ($\theta, \epsilon, \sigma \rightarrow 0^+$) prevents firms from adjusting technical factor proportions, so cost pass-through becomes purely linear and additive, reproducing the classic fixed-price Leontief/Robinson (2006) multiplier.

C. Non-Ricardian demand and disposable-income loops. Standard CGE models enforce intertemporal optimisation in which households anticipate future tax liabilities, neutralising debt-financed spending. Here the consumption block routes household income $E_h = (1 - \tau)w \sum_i L_i +$ capital income; under F3 the absence of an immediate tax hike allows the newly generated wage income $w \cdot \Delta L$ to fuel private consumption directly, generating an endogenous Keynesian multiplier.

E.5 Summary and methodological takeaway

1. **The binary wage regime dominates.** The literature often debates the elasticity of factor substitution or intersectoral mobility (η). As shown in `definitive_guide.md` and our runs, intersectoral mobility produces second-order reallocation effects, whereas the wage regime (flexible ALPHA versus sticky GAMMA) is the first-order determinant of the aggregate multiplier.
2. **Reconciling applied discrepancies.** When policy studies using macroeconomic models (FIDELIO, E3ME) report output multipliers around 1.5 for green investments while standard

CGE models report zero or negative multipliers, the divergence is not an empirical dispute over green technology. It is a direct mathematical consequence of evaluating the shock at GAMMA/DELTA-F3 (slack resources, wage stickiness, debt financing) versus ALPHA-F2 (full employment, market clearing, domestic tax financing).

Table A1: The sectoral family in the `matrix_5x3_v10` generation (`matrix_5x3-v10-BETA-*`; deviations from the no-programme baseline; F2 and F3 agree to all printed decimals in every cell).

Sectoral cell	$\max p - 1 $ (F1)	$\max p - 1 $ (F2)	Employment (F2)	Consumption (F2)
uniform $\eta_s = 0.25$	0.133610	0.152897	1.000864	-0.016888
uniform $\eta_s = 0.5$	0.095839	0.105666	1.001255	-0.015712
uniform $\eta_s = 1$	0.061283	0.065422	1.001627	-0.014669
uniform $\eta_s = 2$	0.035629	0.037171	1.001914	-0.013912
rigid programme, $\eta_s = 0.5$	0.215061	0.272033	0.996621	-0.026914
rigid largest half, $\eta_s = 0.5$	0.109583	0.123362	1.000196	-0.019053
BF ($\eta_{s,i} = 0$)	0.221442	0.279009	1.000000	-0.019800
GAMMA (fixed wage)	0.000000	0.000000	1.001260	-0.015333

Table A2: The equivalence rows in the `matrix_5x3_v10` generation (run ids `matrix_5x3-v10-*`; deviations from the no-programme baseline).

Row	Financing	Real GDP	Consumption (welfare)	Employment L
ALPHA	F1	+0.000000	+0.001432	1.000000
BETA	F1	+0.000000	+0.001432	1.000000
ALPHA	F2	+0.000000	−0.018226	1.000000
BETA	F2	+0.000000	−0.018226	1.000000
ALPHA	F3	+0.000000	−0.018226	1.000000
BETA	F3	+0.000000	−0.018226	1.000000
GAMMA	F1	−0.000973	−0.000806	0.999027
DELTA	F1	−0.000973	−0.000806	0.999027
GAMMA	F2	+0.001260	−0.015333	1.001260
DELTA	F2	+0.001260	−0.015333	1.001260
GAMMA	F3	+0.017796	+0.022644	1.017796
DELTA	F3	+0.017796	+0.022644	1.017796
BF	F1	−0.000110	+0.000963	1.000000
BF	F2	−0.000162	−0.019800	1.000000
BF	F3	−0.000162	−0.019800	1.000000

Labour row	F1 (demand tilt)	F2 (tax-financed)	F3 (debt / external)
ALPHA (full L)	sectoral reallocation	full crowding out	crowding out (w bids up)
BF (immobile)	bottlenecks / wedges	contractionary wedge	bottleneck limits
BETA (elastic)	moderate shift	mild expansion	expansion via w/P
GAMMA (fixed w)	demand-driven shift	balanced-budget multiplier	Keynesian multiplier
DELTA (IO/Leontief)	IO composition shift	Leontief balanced multiplier	pure IO multiplier

Table A3: Where the standard CGE and the FIDELIO model sit in the matrix.